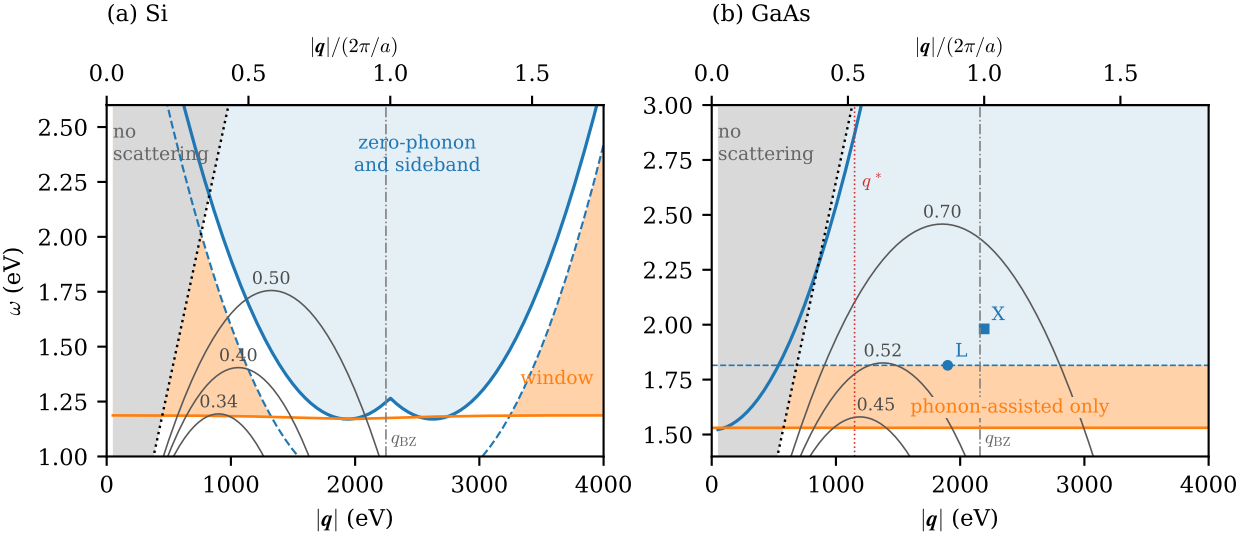

(a) Si
(b) GaAs
$|\boldsymbol{q}|/(2\pi/a)$
0.0
0.5
1.0
1.5
no scattering
zero-phonon and sideband
window
0.50
0.40
0.34
q_{BZ}
ω (eV)
1.00
1.25
1.50
1.75
2.00
2.25
2.50
$|\boldsymbol{q}|$ (eV)
0
1000
2000
3000
4000
3.00
2.75
q^*
0.70
X
L
0.52
0.45
phonon-assisted only